Supplemental Material

Machine-Learned Interatomic Potential for Predictive Simulation of MoS_2 Epitaxy

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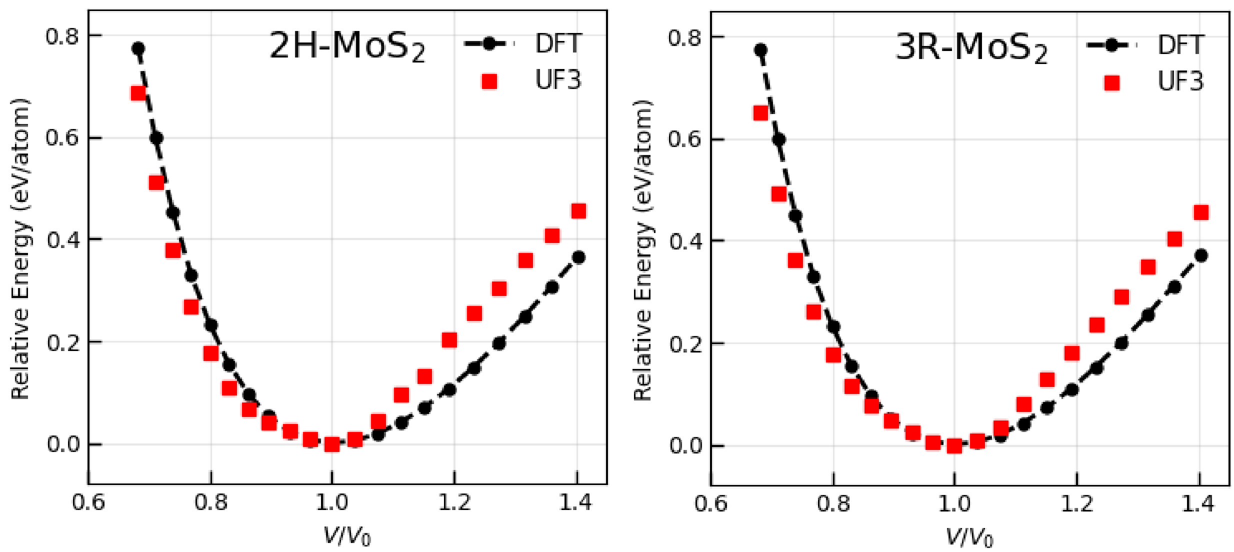

Fig. S1: Relative energy versus volume ratio: UF3 vs. DFT. Due to discrepancies in the predicted cohesive energies of UF3 vs. DFT, we consider the relative energy in comparison. The RMSE values are ~0.069 eV/atom and ~0.066 eV/atom for 2H and 3R, respectively.

Table SI: Fitted Third-order Birch-Murnaghan Equation of State Parameters

	2H		3R	
	DFT	UF3	DFT	UF3
K_0 (GPa)	72.1	81.7	72.8	77.4
K_0'	4.11	4.50	4.10	3.89

**The coefficients were determined by fitting the third-order Birch–Murnaghan functional form[1] to the energy–volume data via nonlinear least-squares regression. The energy as a function of volume was fit to*

$$E(V) = E_0 + \frac{9V_0K_0}{16}\left\{\left[\left(\left(\frac{V_0}{V}\right)^{\frac{2}{3}} - 1\right)^3 K_0'\right] + \left[\left(\left(\frac{V_0}{V}\right)^{\frac{2}{3}} - 1\right)^2 \left(6 - 4\left(\frac{V_0}{V}\right)^{\frac{2}{3}}\right)\right]\right\}$$

where E_0 is the equilibrium energy, V_0 is the equilibrium volume, K_0 is the bulk modulus, and K_0' is its pressure derivative.

Table SII. Energetics of Defect Structures in Monolayer MoS_2: DFT vs. UF3

Defect Structure	**N atoms**	DFT Energy (eV)	DFT ΔE_{defect} (eV)	**DFT $\Delta E^{\%}_{defect}$ (%)**	UF3 Energy (eV)	UF3 ΔE_{defect} (eV)	**UF3 $\Delta E^{\%}_{defect}$ (%)**
Pristine Monolayer	75	-558.7	0.0	**0.0**	-126.1	0.0	**0.0**
V_S	74	-539.3	19.3	**3.6**	-122.4	3.7	**2.9**
Mo_{S_2}	74	-536.2	22.5	**4.2**	-118.3	7.8	**6.2**
S_{2Mo}	76	-534.3	24.4	**4.6**	-120.1	5.9	**4.7**
V_{S_2}	73	-532.4	26.2	**4.9**	-118.8	7.3	**5.8**
V_SV_S	73	-532.3	26.3	**4.9**	-118.7	7.3	**5.8**
V_{Mo}	74	-527.9	30.7	**5.8**	-121.3	4.8	**3.8**
$Schottky_{vS_2}$	72	-514.6	44.1	**8.6**	-114.0	12.0	**9.5**
V_{MoS_3}	71	-512.3	46.3	**9.0**	-113.5	12.6	**10.0**
V_{MoS_6}	68	-492.6	66.0	**13.4**	-105.5	20.6	**16.3**

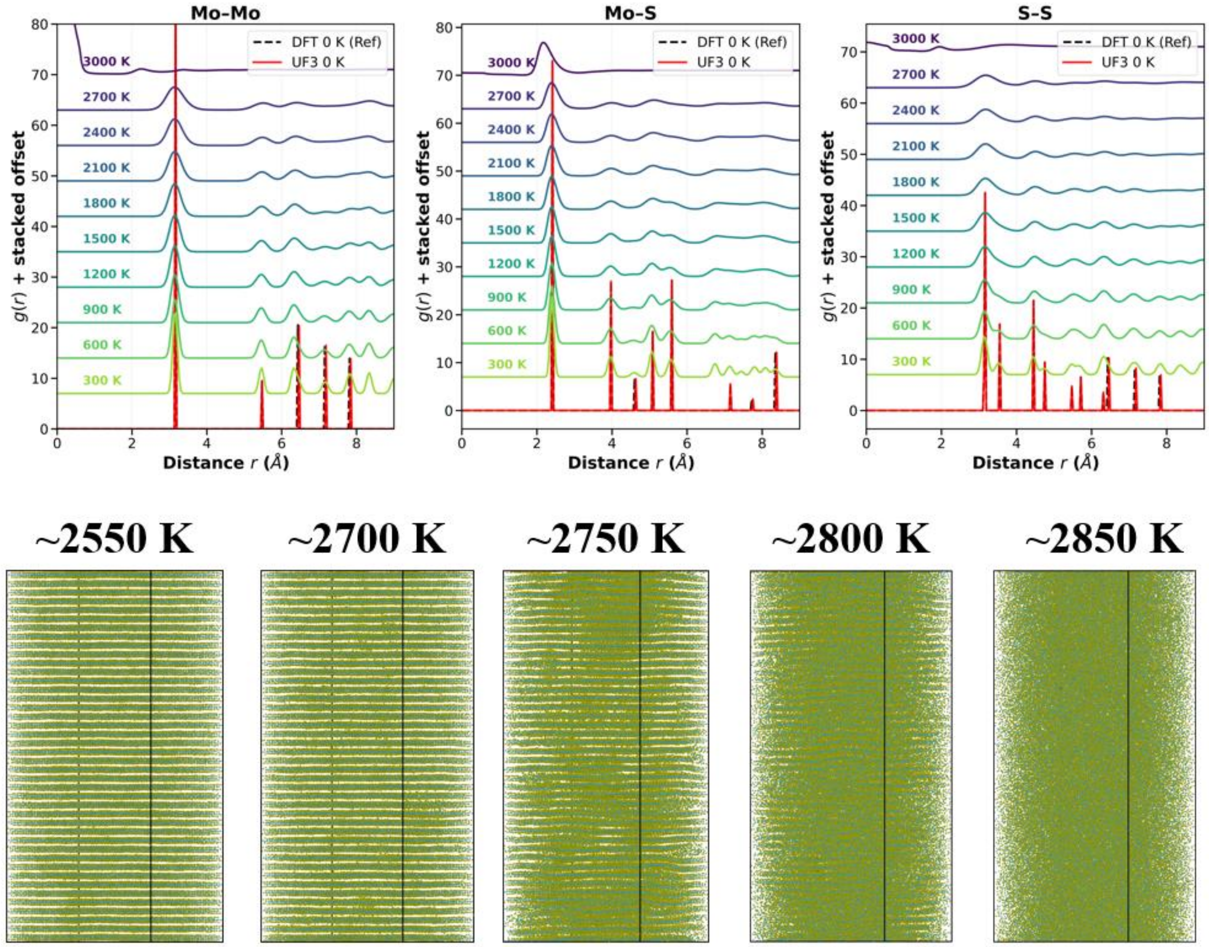

Fig. S2. Temperature-dependent radial distribution functions (RDFs) for bulk 2H-MoS_2. RDFs for Mo–Mo, Mo–S, and S–S atomic pairs are shown for temperatures ranging from 300 to 3000 K (colored curves), with successive temperatures vertically offset by 7.0 for clarity (top panel). Zero-temperature reference RDFs obtained using the UF3 potential (red solid lines) and DFT (black dashed lines) are shown at the baseline. A bulk 2H-MoS_2 system containing 137,214 atoms was heated from 0 K to 3000 K in 300 K increments, with 140 ps of NPT equilibration performed at each temperature (bottom panel). Subsequent 20 ps NVT production runs were performed at every increment and RDF's were time-averaged across the interval. The structure remains stable up temperatures as high as ~2700 K. The results demonstrate that the UF3 potential preserves the structural fidelity of 2H-MoS_2 across a wide temperature range, including temperatures used during the simulation of epitaxial growth (~ 900–1450 K).

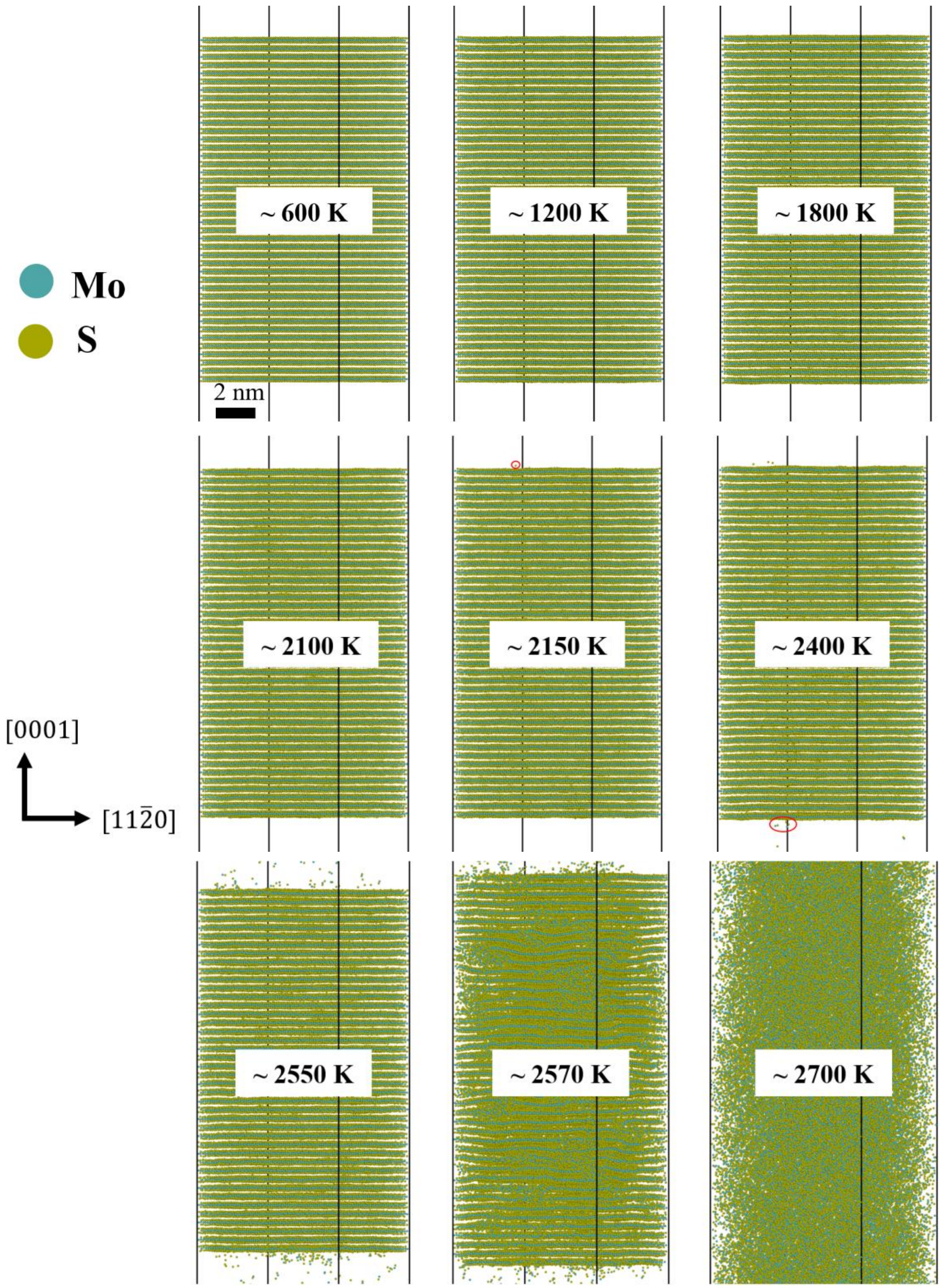

Fig. S3. Thermal stability of a 42-layer 2H-MoS_2 (0001) slab. Atomic snapshots of the 137,214-atom system show structural stability across a wide temperature range, with the onset of surface desorption near 2150 K and structural collapse around 2550 K. The slab was heated from 0–3000 K in 300 K increments using 100 ps NPT ramps with uncoupled in-plane pressure control, followed by 40 ps NPT equilibration and a 20 ps NVT temperature hold at each temperature.

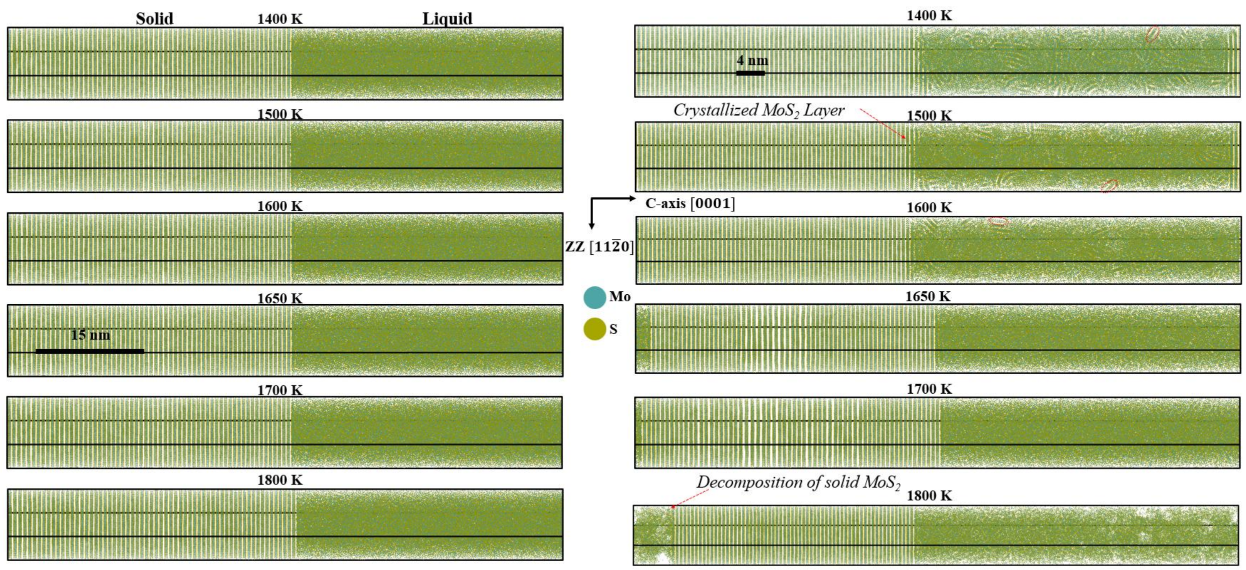

Fig. S4: Two-phase (coexistence) simulations of 2H-MoS_2 using the UF3 MLIP at various target temperatures between 1400 and 1800 K at ambient pressure. The 158,760-atom specimens used in simulation have approximate dimensions of 7 x 6 x 75 nm^3 . The left column shows the initial two-phase structure prior to contact between phases, consisting of crystalline solid and molten regions at the target temperature. The right column shows representative atomic snapshots after equilibration at each target temperature. At T_{guess} = 1400 – 1600 K, crystallization of the molten region is observed. At T_{guess} = 1800 K, the molten phase expands rapidly and the solid MoS_2 decomposes near the interface. At T_{guess} = 1650 – 1700 K, the solid and molten phases coexist without a clear thermodynamic preference, indicating the approximate equilibrium melting temperature, T_m.

Supplementary Note 1: To determine the equilibrium melting/decomposition temperature, T_m, of 2H-MoS_2 predicted by the UF3 MLIP, we employ the two-phase (coexistence) method[2,3]. This method removes the free-energy barrier associated with homogeneous nucleation of solid–liquid interfaces (e.g., those which form during heating and decomposition of a bulk single crystal).

2H-MoS_2 specimens consisting of 158,760-atoms (21x21x60 unit-cells) with approximate dimensions of 7 x 6 x 75 nm^3 are heated to a series of target temperatures, T_{guess} , at zero pressure. The simulation cell is partitioned along the out-of-plane (Z) direction into "solid" and "liquid" regions. The liquid region is heated to ~ 3600 K, well above the expected melting or decomposition temperature, while the single-crystal solid region is maintained at T_{guess}. The liquid region is then rapidly quenched back to T_{guess}, producing a system containing both solid and molten phases separated by a well-defined interface at the target temperature.

The two regions are brought into contact and the system is evolved under an isobaric–isothermal ensemble at ambient pressure. The solid–liquid interface serves as a nucleation site for the growth of either phase. If $T_{guess} < T_m$, the solid phase grows, i.e., the molten phase crystallizes, whereas if $T_{guess} > T_m$, the molten phase grows and the solid melts or decomposes. At $T_{guess} \sim T_m$, the two phases coexist with approximately no net growth of either phase.

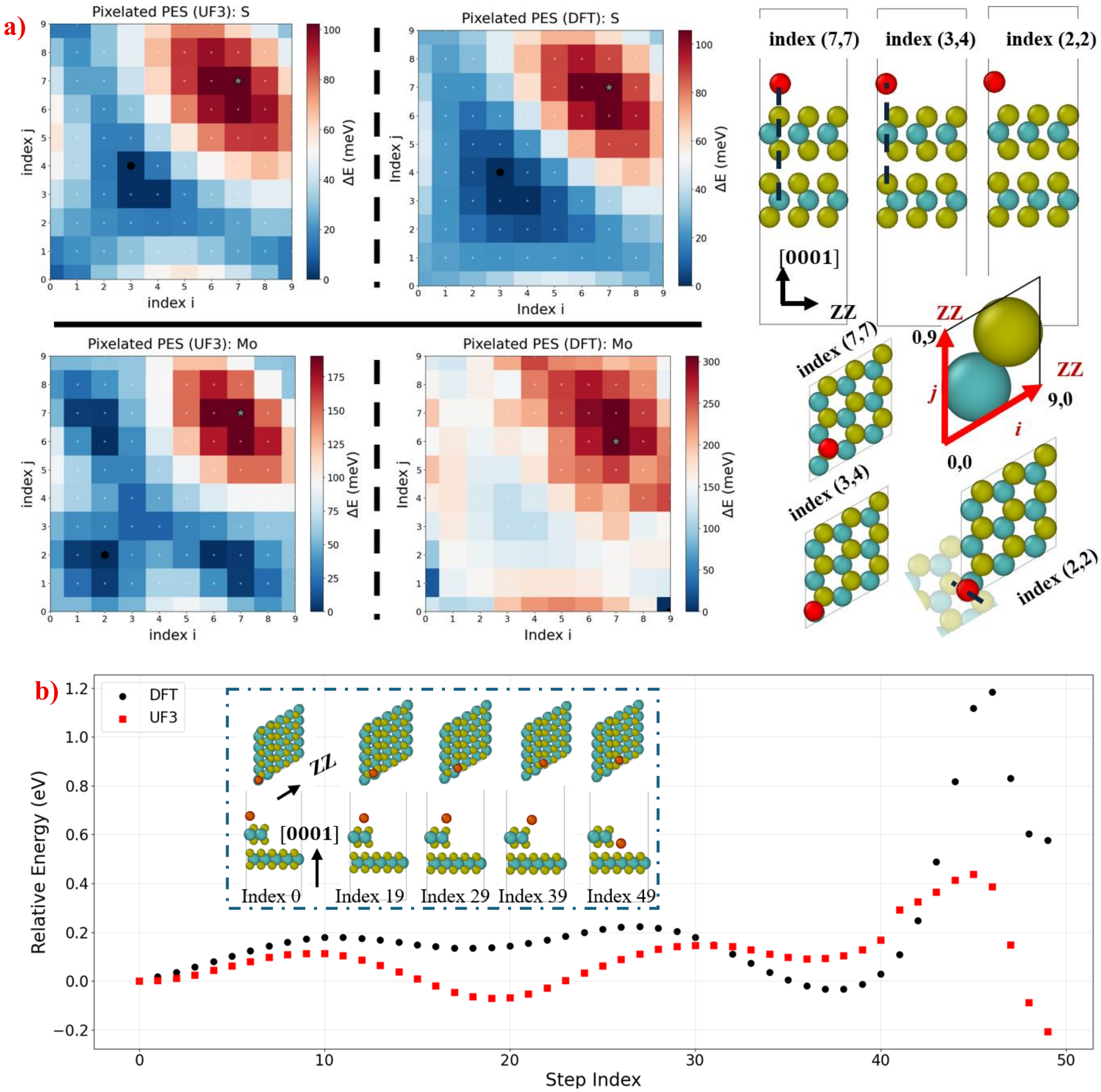

Fig. S5. (a) Sampled potential energy surface (PES) of S (top) and Mo (bottom) adatoms on the basal (0001) surface of 2H-MoS_2 predicted by UF3 (right) and DFT (left). The in-plane coordinates of the adatom were varied on a uniform 10 × 10 grid spanning the lateral area of a single primitive surface unit cell (i.e., one surface translational period). The adatom height was fixed to a vertical displacement equal to the van der Waals gap in bulk 2H-MoS_2, and the total energy of each configuration was normalized relative to the minimum energy sampled across the grid. The indices i and j denote the fractional in-plane coordinates of the adatom along the primitive lattice vectors, as illustrated schematically in the right panel. Each square of the pixelated grid is colored according to the value corresponding to the nearest sampled index pair. The minimum and maximum energy configurations are indicated by a black dot and gray star, respectively. Across the 100 sampled configurations, both UF3 and DFT predict that an S adatom positioned approximately above a Mo lattice site of the underlying layer (index (3,4)) corresponds to the lowest sampled energy, whereas positioning the adatom approximately above an S lattice site (index (7,7)) corresponds to the highest sampled energy. We note that indices (3,4) and (7,7) lie close to the lateral positions occupied by S and Mo atoms, respectively, in the next layer of ideal 2H-MoS_2 stacking, as illustrated in the right panel. The maximum normalized energy sampled for S adatoms is approximately 102 meV in UF3 and 106 meV in DFT. For Mo adatoms, UF3 predicts that index (7,7) corresponds to the highest sampled energy (DFT: (7,6)), while the lowest sampled energy occurs at index (2,2) (DFT: (9,0)). Index (2,2) is slightly offset from the underlying Mo lattice site, positioning the adatom approximately midway between two adjacent S atoms in the topmost layer as shown in the right panel. The maximum normalized energy sampled for Mo adatoms is approximately 190 meV in UF3 and 306 meV in DFT. **(b) Relative energy versus step index for an S adatom descending a zigzag (ZZ) edge along a constrained trajectory predicted by UF3 and DFT**. The adatom was placed at a vertical distance equal to the van der Waals gap above the terrace surface and translated in discrete increments along the in-plane displacement vector connecting the initial and final lateral coordinates. The out-of-plane (Z) coordinate followed the trajectory shown in the inset. A total of 50 configurations (indices 0–49) were uniformly sampled along this trajectory, and the energy of each configuration was normalized relative to that of the initial configuration (index 0). UF3 reproduces some qualitative features of the sampled energy profile relative to DFT, although it underestimates the maximum relative energy along the trajectory (UF3: ~0.44 eV; DFT: ~1.18 eV).

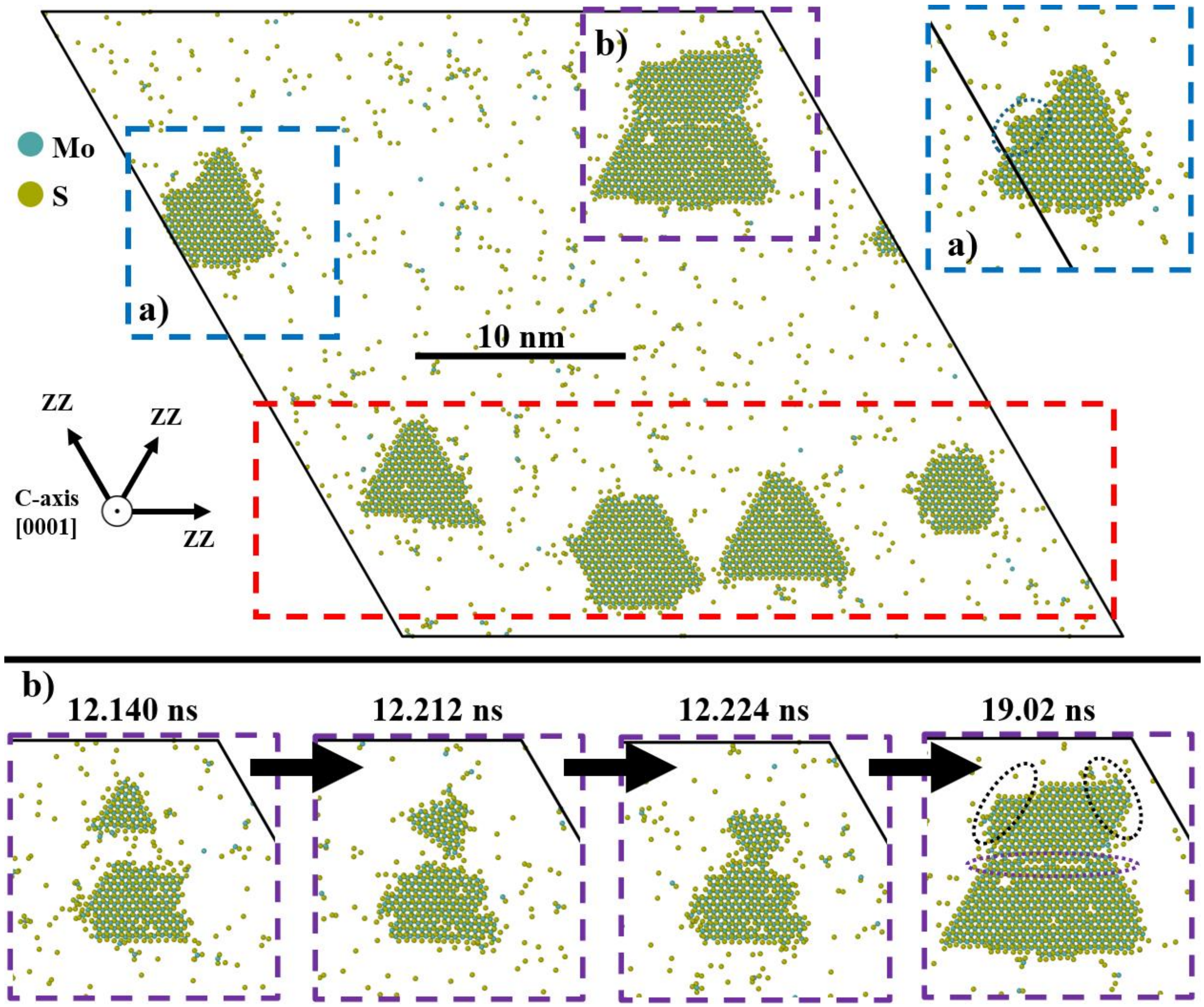

Fig. S6. Nucleation of multiple ZZ-edge-dominated MoS_2 clusters. The simulation was performed on a 12-layer 2H-MoS_2 substrate containing 419,904 atoms (108 × 108 unit cells in-plane) with dimensions of ~34.0 × 29.5 × 7.5 nm^3 at ~1400 K for 19.02 ns. Morphologies exposing zigzag (ZZ) edges dominate, with facets separated by ~60°, consistent with the hexagonal symmetry of the MoS_2 lattice and experimentally observed growth morphologies[4]. For visualization, only the growing epilayer is shown. The dashed red region highlights characteristic triangular and hexagonal-like domains exposing S-terminated ZZ edges. Inset (b) highlights a triangular domain within the dashed blue region, where a smaller triangular subcluster nucleates along one of the ZZ-terminated edges (dotted blue oval). Inset (c) shows time-evolution snapshots of the dashed purple region. A small triangular MoS_2 cluster nucleates, rotates, and becomes misoriented relative to a neighboring growing cluster, forming a boundary between them (dotted purple oval). The morphology of the larger cluster remains dominated by S-terminated ZZ edges with a triangular-like shape containing triangular subclusters (dotted black ovals).

References

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